

Unit 6

Comparing Two Groups

Lecture notes for IB DP Biology and Chemistry

Course purpose	To help students move from raw experimental data to cautious, justified scientific claims.
Unit focus	Independent and paired comparisons, mean differences, confidence intervals, t-tests, effect sizes, assumptions, and non-parametric alternatives.
Central habit	Start from the design and the size of the effect before asking whether a p-value crosses a threshold.

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1 Purpose of Unit 6

Many IB investigations compare two conditions:

- plant growth with and without fertilizer;
- enzyme activity at two temperatures;
- titration results from two analytical methods;
- heart rate before and after exercise;
- bacterial survival in a control group and a treated group.

The statistical question is not simply:

Are the numbers different?

They almost always are. The better question is:

How large is the observed difference, how uncertain is it, and would a difference this large be surprising under a null model of no true difference?

Core thesis

For two-group comparisons, the p-value is never enough. Students should report the observed difference, uncertainty, a justified test, and the biological or chemical meaning of the effect.

By the end of this unit, students should be able to:

1. distinguish independent and paired designs;
2. choose between an independent-samples test and a paired test;
3. calculate and interpret a mean difference;
4. explain why Welch's t-test is often safer than the pooled t-test;
5. explain what a paired t-test actually tests;
6. report effect sizes such as raw mean difference and Cohen's d ;
7. explain when Mann-Whitney, Wilcoxon signed-rank, sign tests, or permutation tests may be useful;
8. avoid false conclusions from $p > 0.05$;
9. write IB-style conclusions that connect statistics to experimental validity.

2 The basic comparison

Suppose a student asks:

Does fertilizer increase plant height after 14 days?

A weak answer would be:

The fertilizer group had a higher mean, so fertilizer worked.

This is not enough. A strong answer needs at least four pieces:

Component	Question it answers
Observed difference	How large was the difference in the sample?
Spread and uncertainty	How variable were the data, and how precise is the estimate?
Statistical test	Would this difference be unusual under a null model of no true difference?
Scientific meaning	Is the effect large enough to matter biologically or chemically?

Common student shortcut

Do not begin with: “Which test should I use?” Begin with: “What is the design, what kind of data do I have, and what claim am I trying to make?”

3 Independent vs paired designs

The first decision is design-based. It is not chosen by the software.

3.1 Independent groups

Use an independent-groups comparison when the individuals in one group are different from the individuals in the other group.

Independent design

Eight plants are grown without fertilizer. Eight different plants are grown with fertilizer. The control plants and fertilizer plants are separate biological units.

In this case, the group means are compared directly:

$$\Delta = \bar{x}_{\text{fertilizer}} - \bar{x}_{\text{control}}$$

3.2 Paired design

Use a paired comparison when each observation in one condition is naturally linked to one observation in the other condition.

Paired design

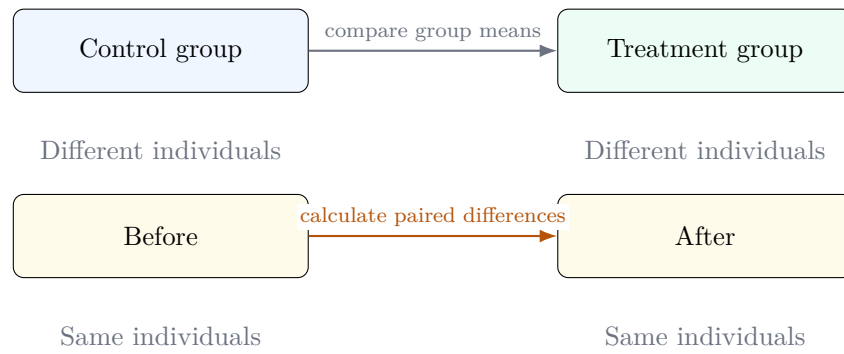
The same student’s heart rate is measured before and after exercise. Each “after” value belongs to a specific “before” value from the same person.

In a paired design, the main data are the differences within each pair:

$$d_i = \text{after}_i - \text{before}_i$$

Then the question becomes:

Is the mean difference $\bar{d}$ clearly different from zero?

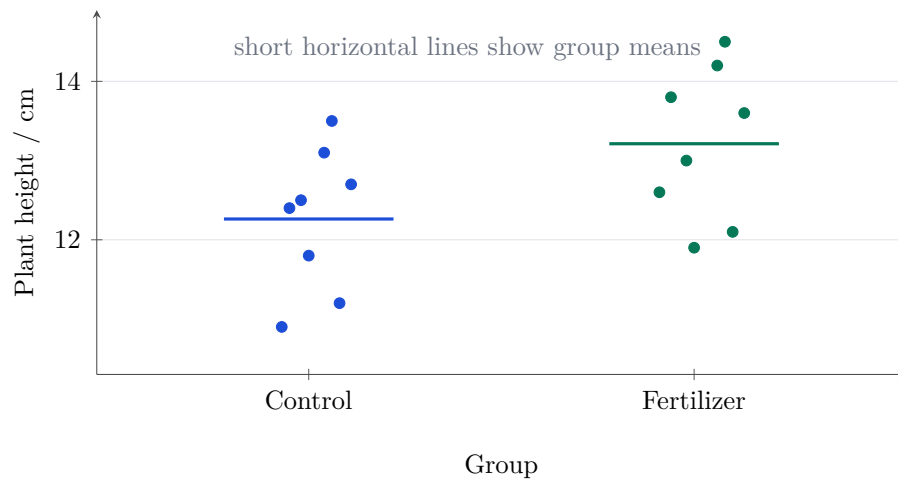


Design rule

If values can be naturally matched pair-by-pair, consider a paired analysis. If the groups contain different independent individuals, use an independent-groups analysis.

4 Raw-data graph first

Before calculating a p-value, plot the raw data. For small datasets, a dot plot is usually better than a bar chart because it shows the individual observations and the overlap between groups.



The graph already tells us several things:

- the fertilizer group has the higher sample mean;
- the groups overlap;
- there is noticeable within-group variation;
- the observed difference needs uncertainty before it can become a strong claim.

5 Mean difference and confidence interval

The simplest effect size is the raw mean difference:

$$\Delta = \bar{x}_1 - \bar{x}_2$$

For the plant example:

Group	n	Mean / cm	SD / cm	SEM / cm
Control	8	12.26	0.90	0.32
Fertilizer	8	13.21	0.96	0.34

The observed difference is:

$$\Delta = 13.21 - 12.26 = 0.95 \text{ cm}$$

This says what happened in the sample. It does not yet say how precisely the population difference is estimated.

For two independent groups, the standard error of the difference is:

$$SE_{\Delta} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

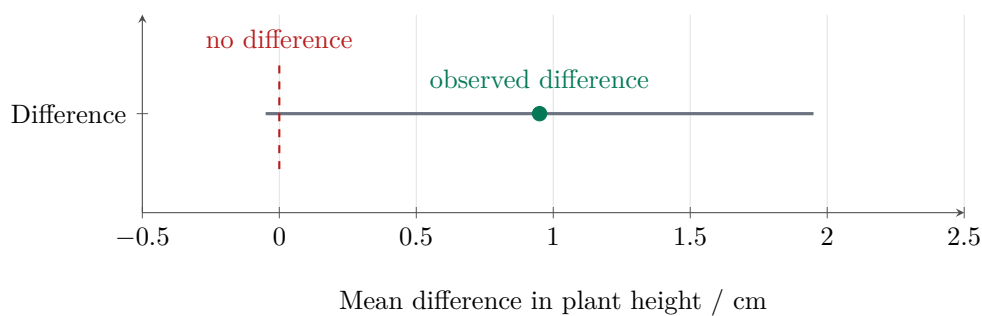
For the example:

$$SE_{\Delta} \approx 0.47 \text{ cm}$$

A 95% confidence interval for the difference is approximately:

$$0.95 \pm 2.15(0.47)$$

$$95\% \text{ CI} \approx -0.05 \text{ to } 1.95 \text{ cm}$$



How to read the interval

The sample suggests fertilizer increased plant height by about 0.95 cm, but the interval is wide and includes values close to zero. The result is suggestive, not decisive.

6 Independent-samples t-test

The independent-samples t-test is used when:

- the outcome is continuous, such as height, mass, reaction rate, or concentration;
- there are two independent groups;
- the aim is to compare group means;
- observations are independent within and between groups.

The statistical hypotheses are usually:

$$H_0 : \mu_1 = \mu_2$$

$$H_A : \mu_1 \neq \mu_2$$

or, equivalently:

$$H_0 : \mu_1 - \mu_2 = 0$$

The test statistic has the general form:

$$t = \frac{\text{observed difference} - \text{null difference}}{\text{standard error of the difference}}$$

For a null difference of zero:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{SE_{\Delta}}$$

Meaning of the t-statistic

A t-statistic measures how many standard errors the observed difference is away from the null difference. A larger absolute t-value means the observed difference is farther from what H_0 expects.

7 Pooled t-test vs Welch's t-test

There are two common versions of the independent-samples t-test.

Version	Assumption	Practical comment
Pooled t-test	The two population variances are equal.	Can be misleading when group spreads or sample sizes differ.
Welch's t-test	Does not assume equal variances.	Usually safer as a default for independent groups.

Welch's standard error is:

$$SE_{\Delta} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

The degrees of freedom are adjusted using an approximation:

$$df \approx \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1-1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2-1}}$$

Students do not need to memorise this formula for IB, but they should understand the reason:

Welch's test accounts for unequal group variability instead of pretending the two groups have the same spread.

Recommended default

For two independent groups with continuous data, Welch's t-test is usually the safer default than the pooled t-test, especially in student experiments where equal variances are rarely guaranteed.

8 Worked example: fertilizer and plant height

Data were collected from two independent groups of plants.

Control / cm	10.9, 12.5, 13.1, 11.2, 12.4, 13.5, 11.8, 12.7
Fertilizer / cm	12.6, 13.0, 14.2, 12.1, 13.8, 14.5, 11.9, 13.6

Summary:

Group	n	Mean / cm	SD / cm
Control	8	12.26	0.90
Fertilizer	8	13.21	0.96

Observed effect:

$$\Delta = 0.95 \text{ cm}$$

Welch's t-test gives approximately:

$$t = 2.03, \quad df \approx 13.94, \quad p \approx 0.061$$

At $\alpha = 0.05$, this is not conventionally significant.

Do not write this

"The fertilizer had no effect because $p > 0.05$."

A better conclusion is:

The fertilizer group had a higher mean plant height by about 0.95 cm. However, the 95% confidence interval was approximately -0.05 to 1.95 cm, and Welch's t-test did not reach the conventional $\alpha = 0.05$ threshold, $p \approx 0.061$. The data suggest a possible positive effect, but the sample is too small and uncertain for a strong conclusion.

This is a good example because it prevents the false habit of treating $p = 0.049$ and $p = 0.051$ as entirely different worlds.

9 Paired t-test

A paired t-test is used when the comparison is within the same units:

- before and after exercise in the same students;
- absorbance before and after treatment in the same sample;
- two methods applied to the same water samples;
- left and right leaves from the same plant;
- matched patient samples.

The paired t-test does not mainly compare two separate group means. It tests the mean of the within-pair differences.

$$d_i = \text{after}_i - \text{before}_i$$

$$H_0 : \mu_d = 0$$

$$t = \frac{\bar{d}}{s_d / \sqrt{n}}$$

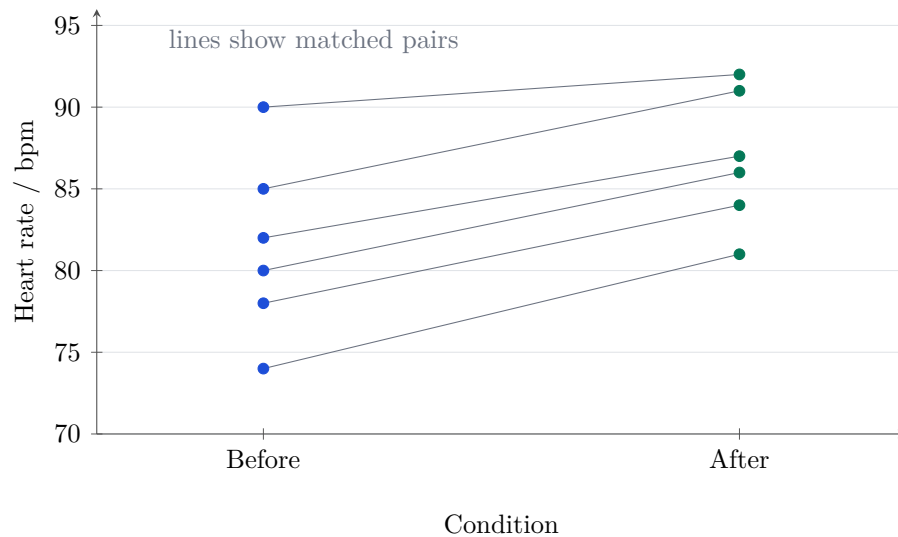
Paired-test intuition

A paired test removes some between-individual variation by asking each subject or sample to act as its own control.

10 Worked example: heart rate before and after exercise

Six students measure their heart rate before and after a short exercise task.

Student	1	2	3	4	5	6
Before / bpm	78	82	74	90	85	80
After / bpm	84	87	81	92	91	86
Difference / bpm	6	5	7	2	6	6



The differences are:

6, 5, 7, 2, 6, 6

$$\bar{d} = 5.33 \text{ bpm}, \quad s_d = 1.75 \text{ bpm}$$

$$t = \frac{5.33}{1.75/\sqrt{6}} \approx 7.46$$

With $df = 5$, this gives approximately:

$$p \approx 0.00068$$

A 95% confidence interval for the mean increase is approximately:

3.50 to 7.17 bpm

A suitable conclusion:

Heart rate increased after exercise in this sample by a mean of about 5.33 bpm. A paired t-test indicated that this increase would be unlikely under a null model of no mean change, $t(5) = 7.46$, $p < 0.001$. The estimated mean increase was approximately 3.50 to 7.17 bpm. The claim is limited by the small sample and short measurement period, but the paired pattern is consistent across most students.

11 Effect size for two groups

A p-value depends strongly on sample size. Effect size describes the size of the difference.

11.1 Raw mean difference

The raw mean difference is often the most important effect size:

$$\Delta = \bar{x}_1 - \bar{x}_2$$

Examples:

- fertilizer increased mean height by 0.95 cm;
- exercise increased heart rate by 5.33 bpm;
- method B gave concentrations 0.12 mol dm^{-3} higher than method A.

This is usually easier to interpret than a standardised effect size because it uses real units.

11.2 Cohen's d

Cohen's d expresses a mean difference in standard deviation units:

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_{\text{pooled}}}$$

where:

$$s_{\text{pooled}} = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

Very rough conventions are:

d	Description
0.2	small
0.5	medium
0.8	large

But these cutoffs should not be used mechanically. In chemistry, a small numerical difference may be very important if it affects safety or analytical validity. In biology, even a small effect may matter if it accumulates across populations or time.

11.3 Hedges' g

Hedges' g is similar to Cohen's d , but includes a small-sample correction. It is often preferred in meta-analysis and small-sample research.

For IB students, the practical rule is:

Report the raw mean difference first. Use Cohen's d or Hedges' g when you need a standardised comparison.

12 Assumptions and diagnostics

The t-test is not just a calculation. It is a model-based method. Its result depends on assumptions.

Assumption	Meaning	How to check or protect
Independence	One observation should not determine another.	Use proper experimental design; avoid treating technical repeats as biological replicates.
Continuous outcome	The outcome is a measurement, not a category count.	Use t-tests for height, mass, rate, concentration, etc.; use categorical methods for counts/proportions.
Approximate normal residuals	Within-group values or paired differences should not be extremely non-normal for small samples.	Plot raw data; inspect outliers; use caution with very small samples.
No extreme outliers	Outliers can dominate means and t-tests.	Investigate mechanism; do not delete inconvenient points without justification.
Similar variances	Required mainly for pooled t-test.	Prefer Welch's t-test when groups have unequal spread.

Normality test trap

A Shapiro-Wilk test with $p > 0.05$ does not prove normality. With small samples it may have low power; with large samples it may detect tiny deviations. Use plots, scientific judgment, and sensitivity checks.

13 Non-parametric alternatives

Non-parametric tests are useful, but they are not a magic repair for weak design or tiny samples.

Situation	Possible alternative	Comment
Two independent groups, strong skew or ordinal data	Mann-Whitney U test	Often described as comparing ranks; not simply a test of means.
Paired data, non-normal differences	Wilcoxon signed-rank test	Uses ranks of paired differences; assumes differences are reasonably symmetric.
Paired data, only direction is reliable	Sign test	Tests whether positive and negative differences are balanced; simple but less powerful.
Small samples with clear exchangeability	Permutation test	Directly simulates the null idea by rearranging labels.

Important caution

If the scientific claim is about means, a non-parametric test may not answer exactly the same question as a t-test. Students should explain what the chosen test is actually comparing.

14 Permutation tests

A permutation test is often the easiest way to understand the logic of a statistical test.

For two independent groups:

1. calculate the observed difference between group means;
2. mix all observations together;
3. randomly reassign them into two groups of the same sizes;
4. calculate the difference in means again;
5. repeat many times;
6. ask how often the random reassignment produces a difference as large as the observed one.

The null idea is:

If treatment has no effect, then the group labels should be exchangeable.

Connection to p-values

A permutation p-value still has the same conditional logic: it estimates how unusual the observed difference would be if the null model of exchangeable labels were true.

Permutation tests are useful for teaching because students can see that the test is not an abstract ritual. It creates a reference distribution under a null model.

15 Choosing the appropriate comparison

Question	Recommended method	Report
Do two independent groups differ in a continuous measurement?	Welch's t-test, with raw-data plot.	Mean difference, CI, p-value, SDs.
Do the same subjects/samples change after treatment?	Paired t-test.	Mean paired difference, CI, p-value.
Two independent groups, strong skew or ordinal scale?	Mann-Whitney U or permutation test.	Median/rank interpretation or permutation logic.
Paired data with non-normal differences?	Wilcoxon signed-rank or sign test.	Median/directional paired change, not just p-value.
Outcome is alive/dead, germinated/not germinated?	Not a t-test. Use binomial, Fisher's exact, or chi-square methods.	Difference in proportions, relative risk, odds ratio.
Several groups?	Not multiple t-tests. Use ANOVA or a planned contrast strategy.	Effect size and multiple-comparison control.

16 IB-style reporting

A poor conclusion says:

The p-value was less than 0.05, so my hypothesis was correct.

A stronger conclusion says:

The treatment group had a higher mean response by Δ units. The confidence interval suggests the plausible size of the effect, and the test indicates whether such a difference would be unusual under the null model. The scientific claim is limited by sample size, measurement uncertainty, and the design.

Weak wording	Better wording
The treatment worked.	The treatment group had a higher mean response under the tested conditions.
There was no difference.	No statistically significant difference was detected.
The t-test proved the hypothesis.	The t-test provided evidence against the null model of equal means.
The result was insignificant, so it does not matter.	The result did not reach the chosen threshold; the effect size and confidence interval should still be considered.
The data were normal because Shapiro-Wilk was not significant.	No strong departure from normality was detected, but sample size and raw-data plots should be considered.

Recommended conclusion structure

For two-group comparisons, write in this order:

1. identify the observed difference in real units;
2. report uncertainty, preferably a confidence interval;
3. report the statistical test and p-value;
4. translate the result into the scientific context;
5. state the main limitations.

17 Practice tasks

Task 1: Identify the design

For each case, decide whether the design is independent or paired.

1. Eight plants receive fertilizer; eight different plants do not.
2. The same water samples are analysed using two titration methods.
3. Heart rate is measured before and after exercise in the same students.
4. Bacterial growth is measured in separate control and antibiotic plates.

Task 2: Choose the method

Choose a suitable analysis and justify it briefly.

1. Two independent groups of plants are compared for final height.
2. The same solution is measured by a colorimeter before and after dilution correction.
3. Survival is recorded as alive/dead in control and treatment groups.
4. Two independent groups have strongly skewed reaction-time data and small sample sizes.

Task 3: Interpret a borderline result

A fertilizer investigation gives:

$$\Delta = 0.95 \text{ cm}, \quad 95\% \text{ CI} = -0.05 \text{ to } 1.95 \text{ cm}, \quad p = 0.061.$$

Write a cautious conclusion. Do not say “no effect.”

Task 4: Paired differences

For paired data, the differences are:

$$4, 5, 3, 6, 2, 5.$$

Calculate the mean difference. Explain what the null hypothesis would be for a paired t-test.

Task 5: Improve the wording

Rewrite the following statement:

The t-test gave $p = 0.12$, proving that the two methods give the same result.

18 Unit 6 checklist

Before choosing a two-group statistical test, students should answer:

1. Are the groups independent or paired?
2. What is the outcome variable, and is it continuous, categorical, ordinal, or a count?
3. What is the observed effect size in real units?
4. How much spread is present within each group?
5. Is the sample size large enough for the intended claim?
6. Is Welch’s t-test more appropriate than a pooled t-test?
7. If paired, have within-pair differences been calculated?
8. Are raw-data plots shown?
9. Is a confidence interval reported?
10. Does the conclusion avoid proof language and “no effect” language?

Final message

Two-group comparison is not a button in software. It is a chain of reasoning: identify the design, plot the raw data, estimate the effect, quantify uncertainty, choose a justified test, and write only the claim that the evidence can support.